# LAB REPORT

**STUDENT NAME**: Boluwatife Adeyeye

**REGISTRATION NUMBE**R: C11.pcdfir2617014

**COURSE CODE**: CIP-B101

**LAB TITLE**: PC System Fundamentals, Disk Sectors and Partition Analysis

**INSTRUCTOR NAME**: Aminu Idris

**DATE**: 05/11/2026

**TOOLS USED:** Windows 10 Home (VM) , Kali Linux v 7.0.12

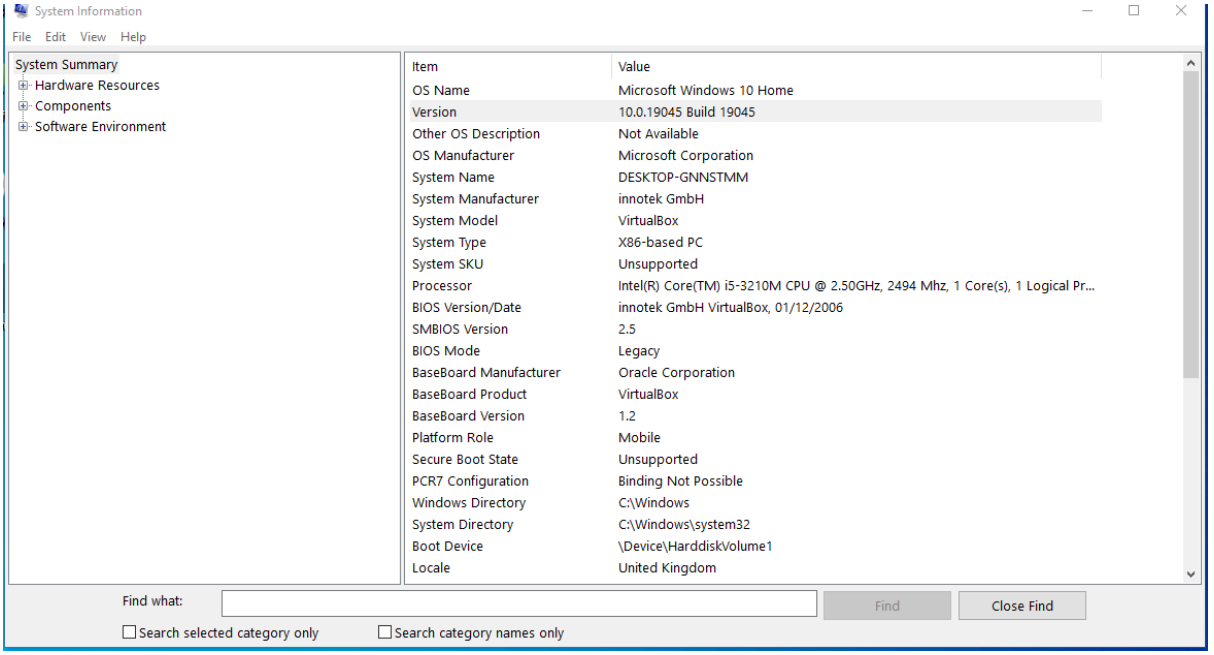
# INTRODUCTION

Disk Sector is the minimum storage unit of a computer hard drive , a drive or disk is divided into sectors and Partition are the ways these sectors are grouped logically and the way they are laid out is recorded into a partition table which maybe Master Boot Record (MBR) or Guid Partition Table(GPT). In Digital forensics  Partition analysis helps an investigator to lactate the beginning and ending of data.

This report aims to examine and observe the basic sector structure of disks and partition layout . The procedure followed , method used and commands used are documented in this report.

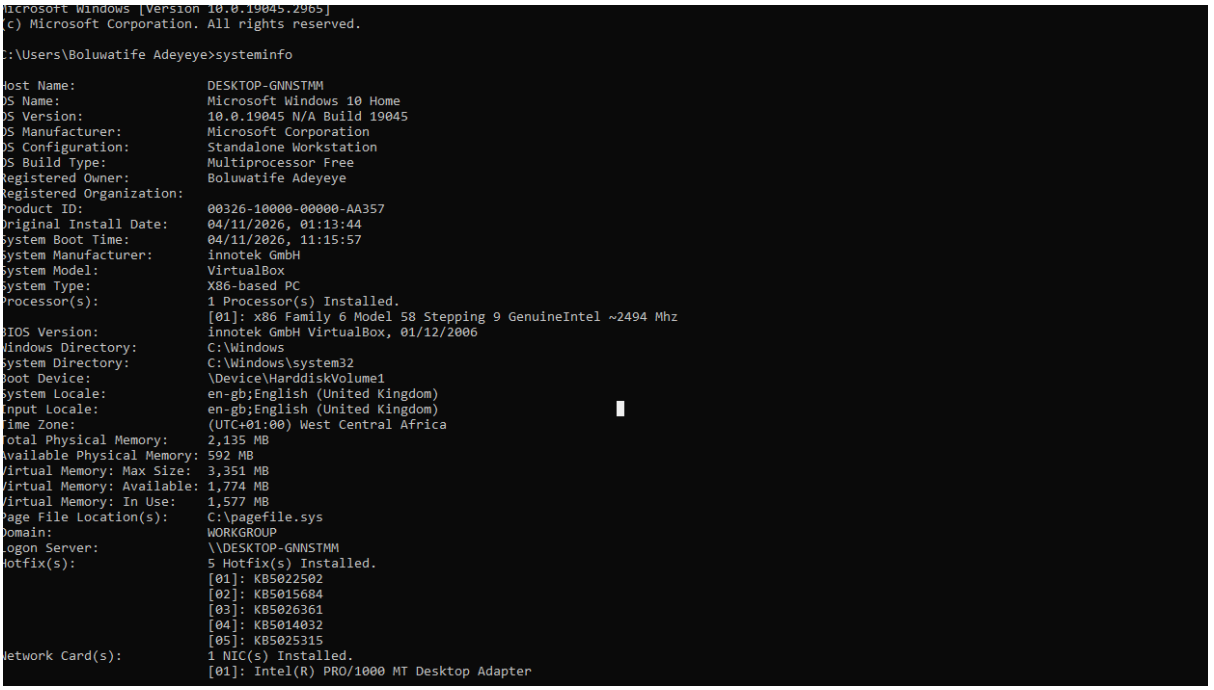
# PART A - System Identification and Evidence Layer Mapping

## A1. Capture basic system specification

- 


Command used: Win + R; msinfo32 .

- Systeminfo;



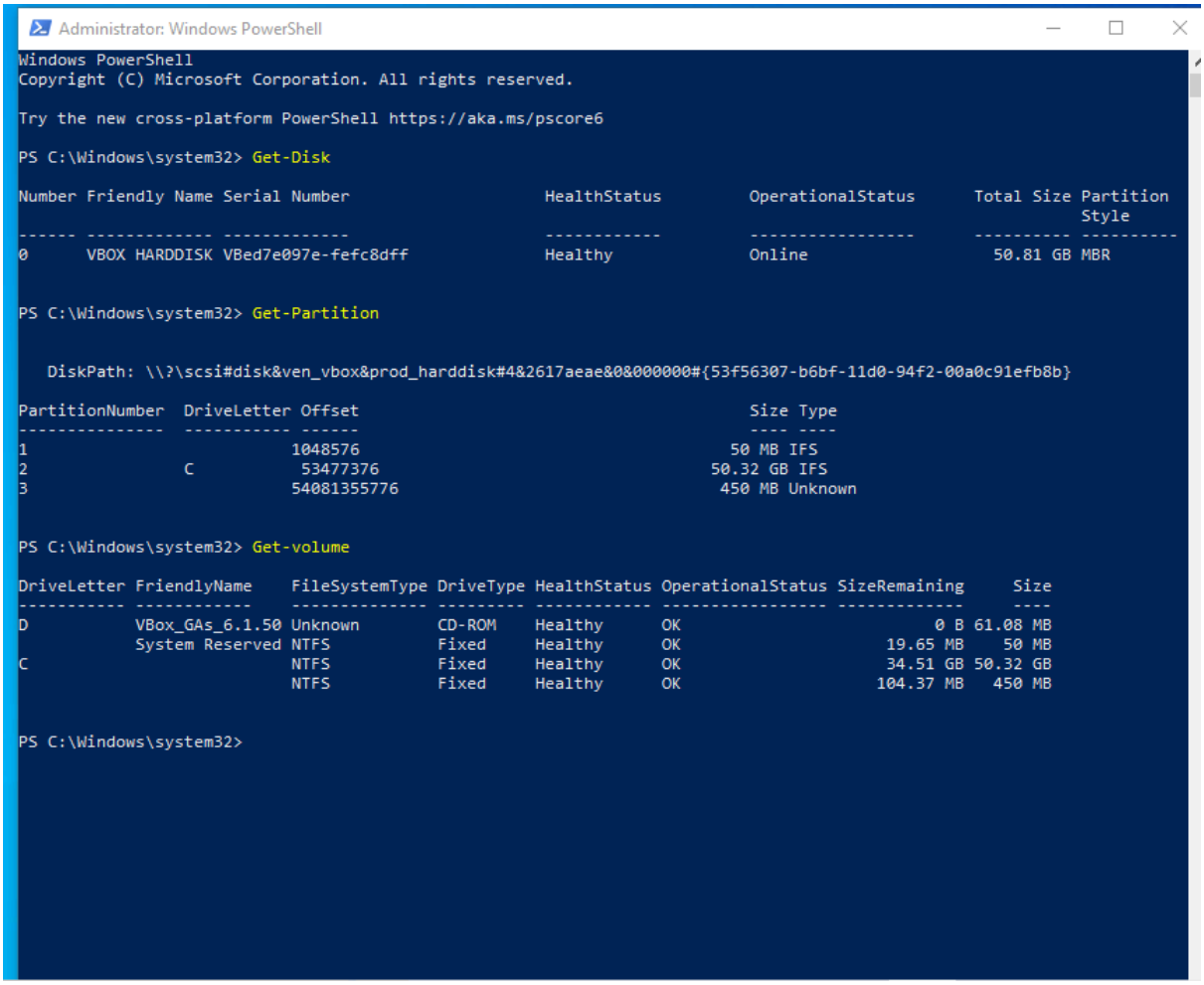
A2. Capture storage Information

- Using Powershell:

Commands; Get-disk

Get-partition

Get-Volume



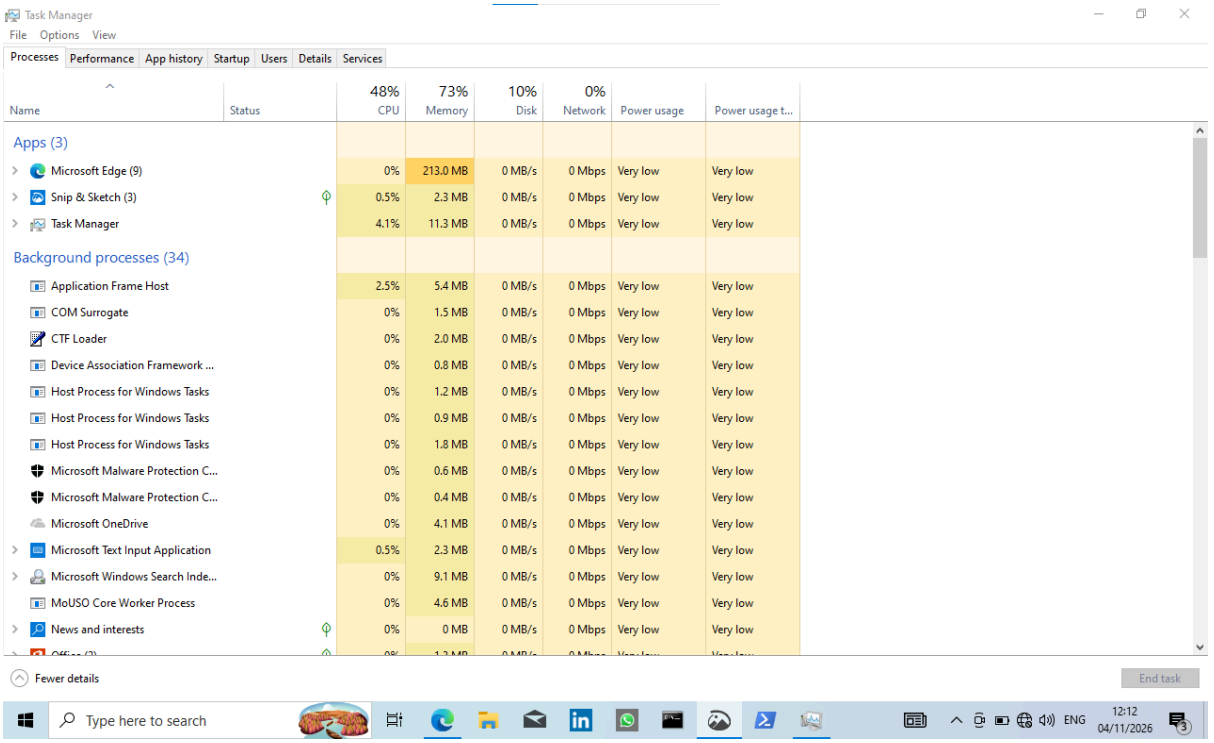
## A3. Map Evidence Sources to System layers

| System layers | Evidence that can be generated from them | |
|---|---|---|
| User Layer | | |
| Application Layer | | |
| Operating System Layer | | |
| Disk Layer | | |
| Memory/ Process Layer | | |
| | | |

# PART B - Program, Process and Thread Investigation

## B1.Observe running processes

- Using Task manager ; showing running processes




- Using Powershell

| Processes Name | PID | Purpose |
|---|---|---|
| ApplicationFrameHost | 5996 | Essential for the proper functioning of Universal Windows Platforms (UWP) apps which are the new types of apps included in windows 10 |
| MpCmdRun | 5748 | It is Microsoft's malware protection command line utility. |
| Microsoft.Photos | 5092 | A windows component for managing and displaying images on a microsofts' computer. |

Command used; Get-Process]Select-object -first 20 Name,id,CPU,workingSet

Difference Between A Program, Process and Thread

A program is an instruction written to be executed by a computer, it is an execution file telling a computer what to do . Process depict a program that is running , so that means for every action a computer is doing ,it represents a program running while Thread is the smallest unit of execution within a process, a process has at least one thread and processes involve many threads and threads share the same resources and memory with a process but run independent sequence of instructions.

Process Information might be useful during malware analysis because it gives an investigator the types of app been used, timestamps and each process has a specific purpose, it may guide in tracking what an attacker has been up to.

## Part C - Disk Sector Size Calculation

### C1. Reproduced slide calculation

Given; sector/track = 400, heads = 12, cylinders = 17000, bytes/sector = 512 and 1 KB = 1024 bytes.

- Total sector = sector/track * heads * cylinders = 400 * 12 * 17000 = 81,600,000 sectors
- Total bytes = Total sector * bytes/sector = 81,600,000 * 512 = 41,779,200,000 bytes
- Kilobytes ; note that 1 KB = 1024 bytes , therefore:

  Kilobytes = total bytes / 1024 = 41,779,200,000 / 1024 = 40,800,000 KB
- Megabytes = total bytes / 1024 * 1024 = 39,843.75 MB
- Gigabytes = total bytes / 1024 * 1024 * 1024 = 38.91GB

| Item | Formula/Value | Answer |
|---|---|---|
| Total sectors | 400*12*17000 | 81,600,000 sectors |
| Total Bytes | 81,600,000*512 | 41,779,200,000 bytes |
| Kilobytes | 41,779,200,000 / 1024 | 40,800,000 KB |
| Megabytes | 40,800,000 / 1024 | 39,843.75 MB |
| Gigabytes | 39,843.75 / 1024 | 38.91 GB |

### C2. Calculating my own disk / partition size

- Partition name size

  Partition 1 50.0 MB

  Partition 2 50.32 GB

Partition 3 450.0 MB

- ( A windows VM whose virtualBox disk = 512 bytes is being used)

1. Using partition 1; offset = 1048576 size= 50 MB Total byte = 50 * 1024 * 1024 = 52,428,800 bytes

2. Using partition 2; offset - 53477776 size = 50.32 GB Total byte = 50*1024*1024*1024 = 52,428,800 bytes

Start sector = offset / 512 = 1048576 / 512 = 2048 bytes

Number of sector = total bytes / 512 = 52,428,800 / 512 = 102,400 sectors

Disk partition size = number of sectors * bytes /sector

Sector size in GB = total bytes/ 1024 * 1024 * 1024 = 0.049 GB

3. Using Partition 2; offset = 53477376, 50.32 GB = 50.32 * 1024 * 1024 * 1024 =Total bytes = 54,027,878,400 bytes

Start sector = offset / 512 = 53477376 = 104,448 bytes

Number of sector = total bytes / 512 = 54,027,878,400 / 512 = 105,523,200 sectors

Sector size in GB = 54,556,822,077 / 1024 * 1024 * 1024 = 50.32 GB

4. Using Partition 3; offset = 54081355776 450 MB = 450 * 1024 * 1024 total byte = 471,859,200 bytes

Start sector = offset / 512 = 54081355776/512 = 105,627,648 bytes

Number of sector = total bytes / 512 = 471,859,200 / 512 = 921,600 sector

Sector size in GB = 471,859,200 / 1024*1024*1024 = 0.44GB

- Screenshot of where the data is from; The file system for the 3 partitions is NFTS

```
PS C:\Users\Boluwatife Adeyeye> Get-disk

Number Friendly Name Serial Number                  HealthStatus         OperationalStatus      Total Size Partitio
                                                                                                           Style
------ ------------- -------------                  ------------         -----------------      ---------- ------
0      VBOX HARDDISK VBed7e097e-fefc8dff            Healthy              Online                   50.81 GB MBR


PS C:\Users\Boluwatife Adeyeye> Get-partition


   DiskPath: \\?\scsi#disk&ven_vbox&prod_harddisk#4&2617aeae&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                     50 MB IFS
2                C           53477376                                 50.32 GB IFS
3                            54081355776                                450 MB Unknown

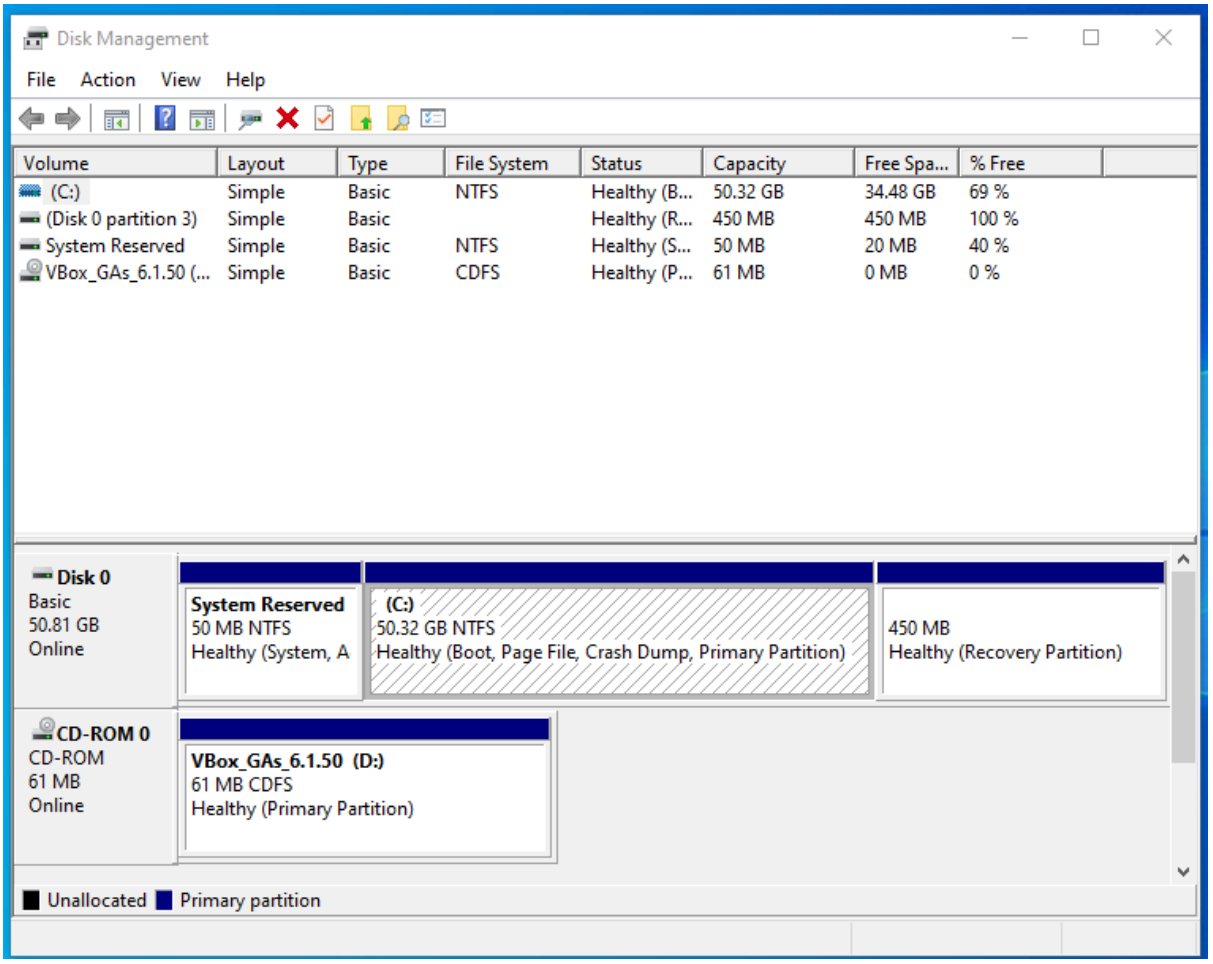

PS C:\Users\Boluwatife Adeyeye> Get-volume

DriveLetter FriendlyName    FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining     Size
----------- ------------    -------------- --------- ------------ ----------------- -------------     ----
D           VBox_GAs_6.1.50 Unknown        CD-ROM    Healthy      OK                        0 B 61.08 MB
            System Reserved NTFS           Fixed     Healthy      OK                   19.65 MB    50 MB
C                           NTFS           Fixed     Healthy      OK                   33.84 GB 50.32 GB
                            NTFS           Fixed     Healthy      OK                  104.37 MB   450 MB
```

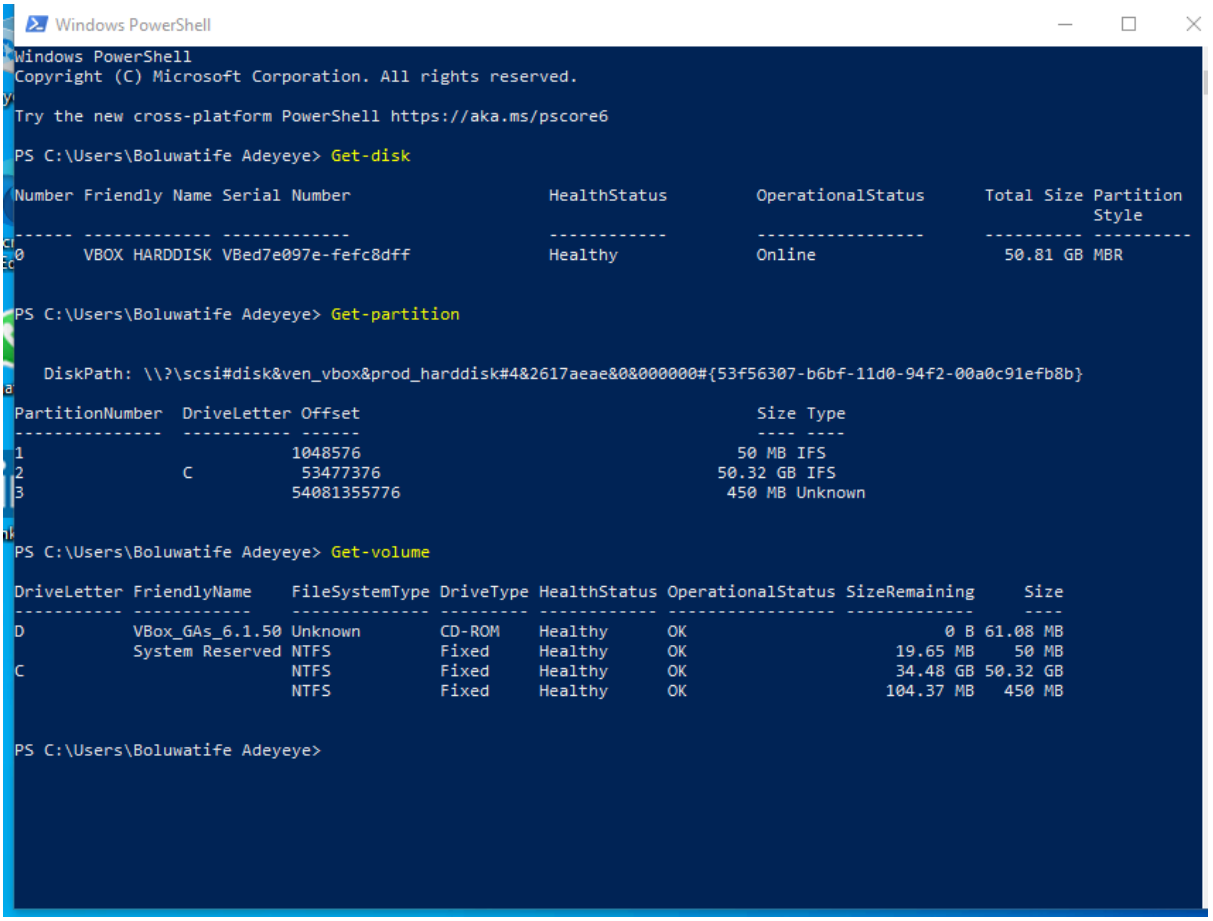
# PART D - Safe Partition Table Inspection

## D1. Windows method - Disk Management, Powershell and DiskPart

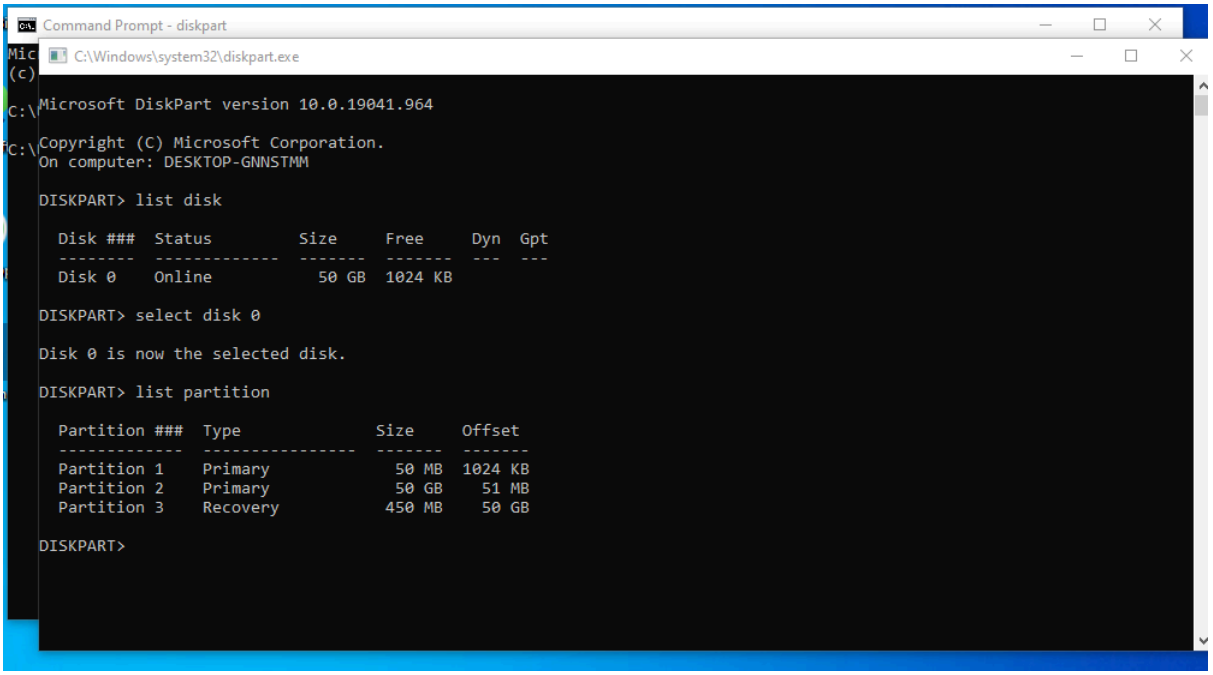
- Disk management -  command ; win + R,  disk mgmt msc



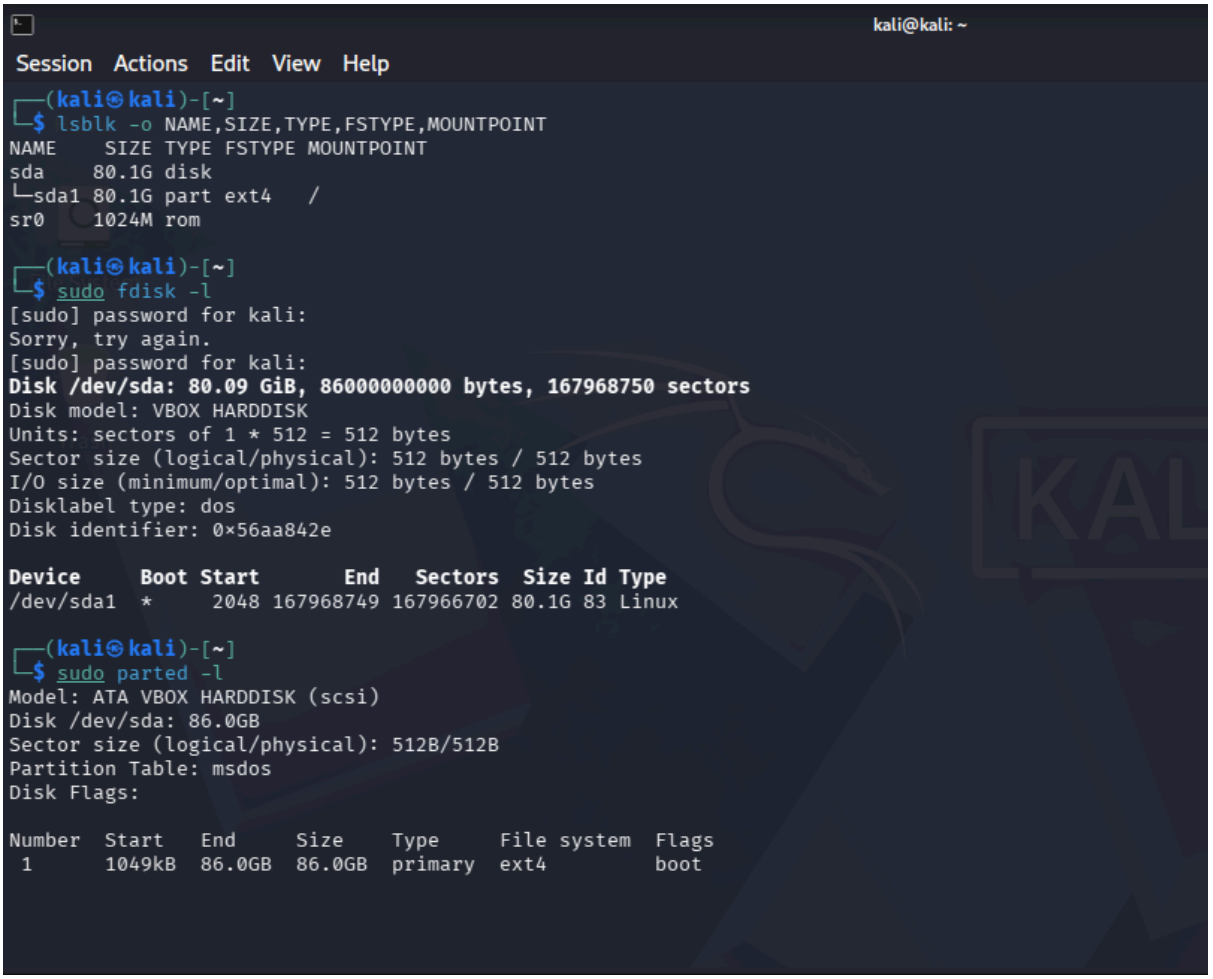

- Disk layout and visible partitions

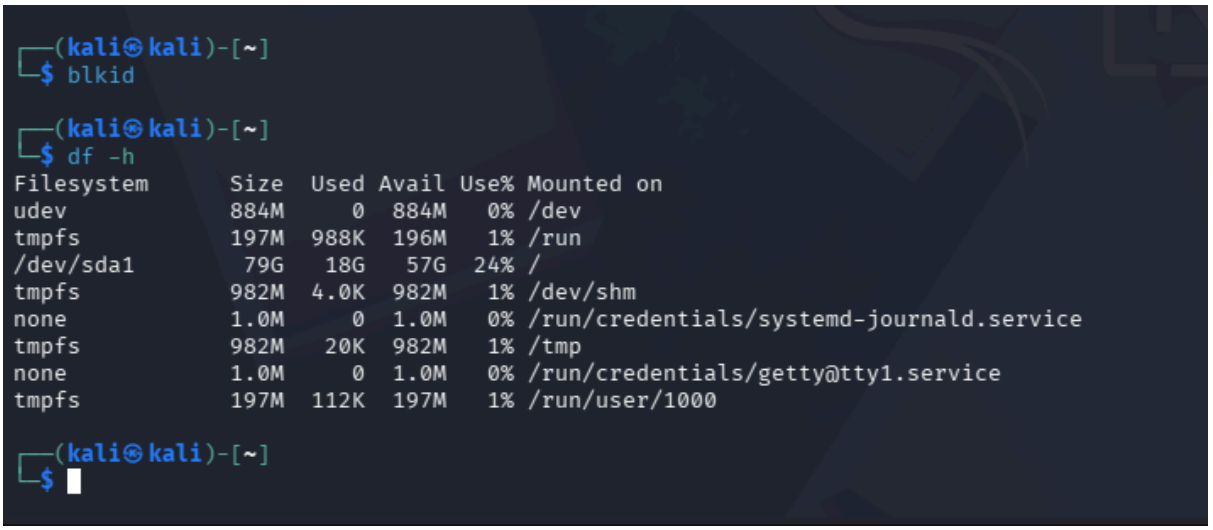



- Disk part listings and partitions




D2. Linux / Kali method - read-only inspection


kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT
NAME    SIZE TYPE FSTYPE MOUNTPOINT
sda    80.1G disk
└─sda1 80.1G part ext4   /
sr0    1024M rom
┌──(kali㉿kali)-[~]
└─$ sudo fdisk -l
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
Disk /dev/sda: 80.09 GiB, 86000000000 bytes, 167968750 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x56aa842e
Device     Boot Start       End   Sectors  Size Id Type
/dev/sda1  *     2048 167968749 167966702 80.1G 83 Linux
┌──(kali㉿kali)-[~]
└─$ sudo parted -l
Model: ATA VBOX HARDDISK (scsi)
Disk /dev/sda: 86.0GB
Sector size (logical/physical): 512B/512B
Partition Table: msdos
Disk Flags:
Number  Start   End     Size    Type     File system  Flags
 1      1049kB  86.0GB  86.0GB  primary  ext4         boot


┌──(kali㉿kali)-[~]
└─$ blkid
┌──(kali㉿kali)-[~]
└─$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            884M     0  884M   0% /dev
tmpfs           197M  988K  196M   1% /run
/dev/sda1        79G   18G   57G  24% /
tmpfs           982M  4.0K  982M   1% /dev/shm
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           982M   20K  982M   1% /tmp
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           197M  112K  197M   1% /run/user/1000
┌──(kali㉿kali)-[~]
└─$

## PART E - MBR, GPT and Boot Process Explanation

### E1. Explained Boot Process

1. Power On: When the power button of a PC or a Computer is pressed, the power supply unit located in the internals of the system sends a power signal to the central processing unit and the CPU resets and starts executing code at a fixed address where the firmware (BIOS, UEFI) is mapped.

2. Firmware starts (BIOS or UEFI): The firmware is a software that runs first, it initialises the hardware like the CPU , memory controller and chipset. The BIOS means Boot

3. Power-On Self-Test (POST): The BIOS/UEFI checks that the essential hardware is working. In a case it fails , the failure stops the boot and is signalled by beep codes or an on-screen error.

4. Complimentary metal-oxide semiconductor(CMOS)/firmware settings: The firmware reads its saved configuration, boot order, date, time and hardware options. On older BIOS systems, these live in batteries backed by CMOS memory.

5. Boot device and Boot records : BIOS is legacy which means it loads the first sector of the boot disk and the MBR. Whereby the MBR code finds the active partitions and runs its boot sector. For the modern UEFI, it reads the GPT and loads a .efi boot file directly from the EFI system partition (ESP) with no MBR boot code involved.

6. Boot loader/ Boot manager: On windows , this bootmgr reads the Boot configuration data (BCD), shows the OS menu if any and starts winload which is equivalent to GRUB on Linux.

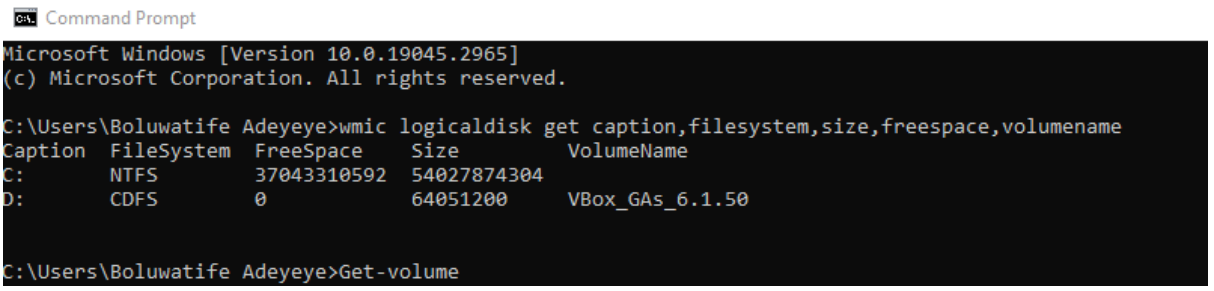
7. Kernel loading: Winload loads the windows kernel, the hardware abstraction layer, the system registry hive and boot-start drivers into memory.

8. Operating system control : The Kernel initialises , starts the session manager , services and login screen and takes full control of the hardware. At this stage , the firmware job is finished.

- My own system uses BIOS /legacy

## PART F - File System Identification and Interpretation

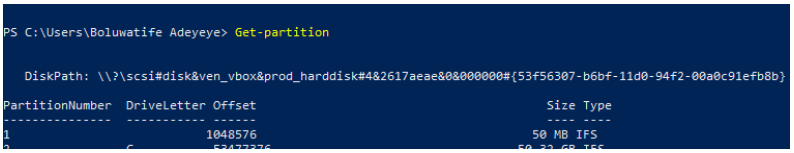
- Partition 1 and 2 on my system uses the NFTS file system.

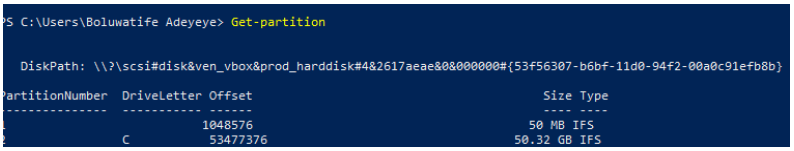



- The VBox_GAs-6.1.50 volume on my system uses another file system which is CDFS (Compact Disk File System) which is designed for read-only and write-once CD-ROMs.

- My linux system uses
- A file system is important in digital forensics because it gives a direction of how to retrieve important files in any storage device which may serve as evidence for investigation.

PART G - Final Practical Assignment

| TASK | Required Evidence | Student Note/ Answer |
|---|---|---|
| Identify partition 1 |  | The partition is on disk o and it is named partition 1. This shows the offset, size and type . |
| Record partition 1 sector count | Start sector = offset / 512 = 1048576 / 512 = 2048<br>Sector count = | The offset is used in calculating the start count by multiplying by the fixed byte size of 512 |
| Calculate partition 1 size | Size = 50 MB<br>Total bytes = 50 * 1024 * 1024 = total byte = 52,428,800 bytes | The size in gigabyte is gotten by dividing with 1024^2 |

| | | |
|---|---|---|
| | size in GB = total bytes/ 1024 * 1024 * 1024 = 52,428,800/1024^3 = 0.049 GB | |
| Identify Partition 2 |  | The partition 2 is of type IFS, size 50.32GB, the largest of the partitions and offset of 5347776 |
| Record partition 2 sector count | Start sector = offset / 512 = 53477376 = 104,448 | Start sector is gotten by multiplying offset by 512 |
| Calculate partition 2 size | Size = total bytes/1024^3<br>Total bytes = 54,556,822,077<br><br>size in GB = 54,556,822,077 / 1024 * 1024 * 1024 = 50.32 GB | Total size in Gigabytes has been given already in the command output |
| Compare OS-reported size | The OS reported size is the same thing also showing the sizes of other partitions | It shows that different commands can be used to achieve the same result. |
| Explain difference if any | None | |